
At the beginning of a season the proportion of infected patches is p and population density is N .

We assume that the population growth follows a logistic model:

$$N \rightarrow N(1 + r - \frac{N}{K})$$

Then we have the epidemic. After this season P_1p of the originally infected patches persists. We assume that during the season $Bp(1-p)$ of the patches get infected, where B is a constant. So after the season p becomes $P_1p + Bp(1-p)$

And a proportion of $zDP_{nf}p$ of the population dies because of the epidemic, P_{nf} is the non-fizzle probability.

So the overall function is

$$f : \begin{pmatrix} p \\ N \end{pmatrix} \rightarrow \begin{pmatrix} P_1p + Bp(1-p) \\ (1 - zDP_{nf}p)(1 + r - \frac{rN}{K})N \end{pmatrix}$$

The disease-free equilibrium is $p = 0$ and

$$N_{DFE} = K$$

Local linearization at DFE is

$$J_f = \begin{pmatrix} B + P_1|_{N=K} & 0 \\ -zDP_{nf} & 1 - r \end{pmatrix}$$

So the persistence criterion is

$$L_0 > 1$$

where

$$L_0 = P_1|_{N=K} + B$$

B is a constant between 0 and 1, so L_0 is non-monotonous with R_0 and the decreasing interval is the same with P_1 (shown in figure 5 in burnout paper)

We can say that at the beginning, L_0 is slightly greater than 1, and (ε, R_0) is in the decreasing interval, so the wild-type plague can persist. But there is a decrease in B for some reason and makes $L_0 < 1$. Then decreasing R_0 makes P_1 larger and $L_0 > 1$ again.